now will finish up this very long lecture by showing you what these principles have to do with **X** and **P**.

We'll start with the connection between commutators and classical physics. As you may recall, we found that commutators have a great similarity to Poisson brackets, a relationship we made explicit in Eq. 4.21. If we plug in the operator symbols **L** and **M** that we've been using in this lecture, we get

$$[\mathbf{L}, \mathbf{M}] \qquad \Longleftrightarrow \qquad i\hbar\{L, M\}, \tag{8.26}$$

and we're reminded that the equations for quantum motion strongly resemble their classical equivalents. This suggests that we may learn something by computing the commutator of the observables **X** and **P**. Fortunately, this is easy to do.

First, let's see what the product **XP** does when it acts as an operator on an arbitrary wave function $\psi(x)$. Recalling Eqs. 8.5 and 8.15, we can write

$$\mathbf{X}\psi(x) = \qquad x\psi(x)$$

$$\mathbf{P}\psi(x) = \quad -i\hbar\frac{d\psi(x)}{dx}.$$

Together, these equations tell us how the product **XP** acts on $\psi(x)$:

$$\mathbf{XP}\psi(x) = -i\hbar x\frac{d\psi(x)}{dx} \tag{8.27}$$

Now, let's try it with **X** and **P** in the opposite order: